🛠️ Warehouse KPI Agent - Developer Guide

Complete guide for developers to understand, setup, and contribute to the Warehouse KPI Agent

---

📚 Table of Contents

🏗️ Architecture Overview

System Architecture

The Warehouse KPI Agent is built as a **stateful AI agent** using:

- LangGraph for orchestration
- Ollama for local LLM inference
- MongoDB for data storage

Core Flow:

User Interfaces → LangGraph State Machine → Tools Layer → Data Layer

---

Key Architectural Principles

1. Schema-Driven Design

All schemas live in a **single source of truth**: collections_schema.py

Benefits:

- No scattered logic
- Self-documenting
- Easy updates
- Type-safe queries

---

2. Keyword-Based Routing

- 80+ domain keywords per collection
- 5-layer guardrail system ensures accuracy

---

3. Two-Stage LLM Pipeline

Stage 1: Intent Classification

- Extracts intent + entities

Stage 2: Response Formatting

- Converts DB results → human-readable output

4. Stateful Conversations

- SQLite checkpointing
- Thread-based memory
- Supports follow-ups

💻 Technology Stack

Core

- Python 3.12+
- LangGraph
- LangChain

LLM

- Ollama
- qwen2.5:7b

Data

- MongoDB
- PyMongo
- Pandas
- SQLite

APIs & UI

- FastAPI
- Uvicorn
- Streamlit

Export

- openpyxl
- Rich
- Pydantic

📁 Project Structure

Root

```
warehouse_AI/
├── collections_schema.py
├── kpi_registry.py
├── output/
├── README.md
├── DEVELOPER_GUIDE.md
└── warehouse_kpi_agent/
```

Main Package

```
warehouse_kpi_agent/
├── app/
├── data_ingestion/
├── tests/
```

```
├── requirements.txt
├── Makefile
└── .env
```

Key Directories

app/

- config → settings & schema
- db → database logic
- graph → LangGraph workflow
- services → business logic
- tools → collection queries

data_ingestion/

- raw_data → source CSVs
- pruned → cleaned data

tests/

- intent tests
- KPI tests
- graph flow tests

🚀 Development Setup

Steps

1. Clone repo
2. Create virtual env

```
python3 -m venv venv
source venv/bin/activate
```

3. Install dependencies

```
make install
```

4. Configure .env
5. Start MongoDB

```
make backend
```

6. Setup Ollama

```
ollama pull qwen2.5:7b
```

7. Load data

```
make load-data
```

8. Run tests

```
make test
```

9. Start app

```
make run
```

🔄 Development Workflow

Daily Flow

```
make backend
make run
make test
```

Adding a New KPI

1. Update kpi_registry.py
2. Implement logic in tool
3. Add to KPI list
4. Test

Adding New Collection

1. Add schema
2. Create tool
3. Add keywords
4. Register tool
5. Load data

🧪 Testing Guide

Run Tests

make test
python tests/run_tests.py

Categories

- Intent classification
- KPI calculations
- Graph flow

🤝 Contributing

Code Style

- Follow PEP 8
- Use type hints
- Write docstrings

Commit Format

feat: add KPI
fix: correct calculation
docs: update guide

🔧 Troubleshooting

MongoDB Issues

make backend-stop
make backend

Ollama Issues

ollama list
curl http://localhost:11434/api/tags

Test Failures
make load-data
make test

Import Errors
source venv/bin/activate

📚 Quick Reference

Commands
make setup
make backend
make load-data
make test
make run
make api
make ui

Key Files

Task	Location
Add KPI	kpi_registry.py
Schema	collections_schema.py
Routing	graph/conditions.py
Keywords	intent_classifier.py
Exports	excel_exporter.py

Environment Variables
- MONGODB_URI
- MONGODB_DB
- OLLAMA_BASE_URL
- LLM_MODEL
- LOG_LEVEL

🚀 Ready to Start
Run:
make setup
and begin exploring the project!